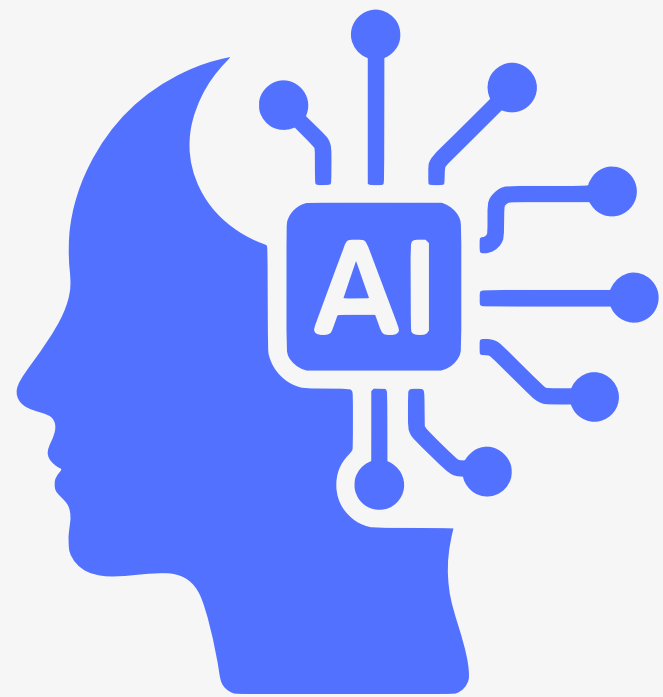




CACHING & RATE LIMITING

WHAT IF ONE VIRAL AI APP COULD
BANKRUPT YOUR STARTUP IN A SINGLE DAY?





WHAT IS CACHING?

- Stores frequently used data temporarily
- Avoids repeated API calls
- Improves response speed
- Reduces server workload
- Saves AI API costs





REDIS CACHING

- In-memory caching database
- Extremely fast
- Stores temporary data
- Supports expiration (TTL)
- Widely used in production systems





AI COST REDUCTION

- Avoid duplicate AI requests
- Cache repeated prompts
- Optimize token usage
- Reduce API calls
- Lower infrastructure costs





API PROTECTION

- Limits requests per user
- Prevents API abuse
- Protects against spam
- Reduces server overload
- Improves application security





SCALING BASICS

- Handle more users efficiently
- Optimize performance
- Reduce response time
- Distribute workload
- Build reliable AI systems

